Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 16 GB of memory. The dataset was split into 80 percent training data and the rest for testing.

2 Results

Figure 1 shows the training loss.

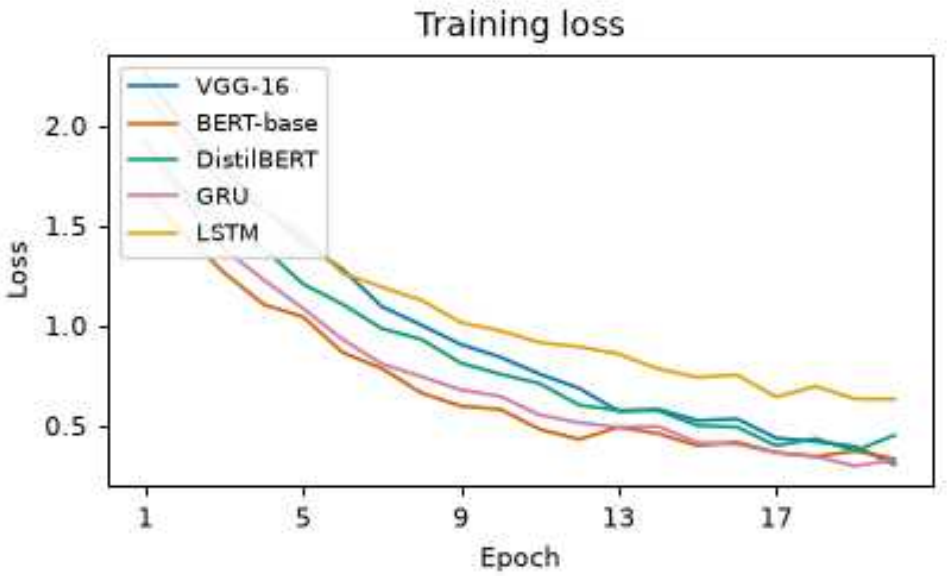


Figure 1: Training loss for the models in this experiment.

VGG-16 has a loss of 1.499 at epoch 18. At epoch 3, the loss of GRU is 1.488.

Figure 2 shows the accuracy by dataset.

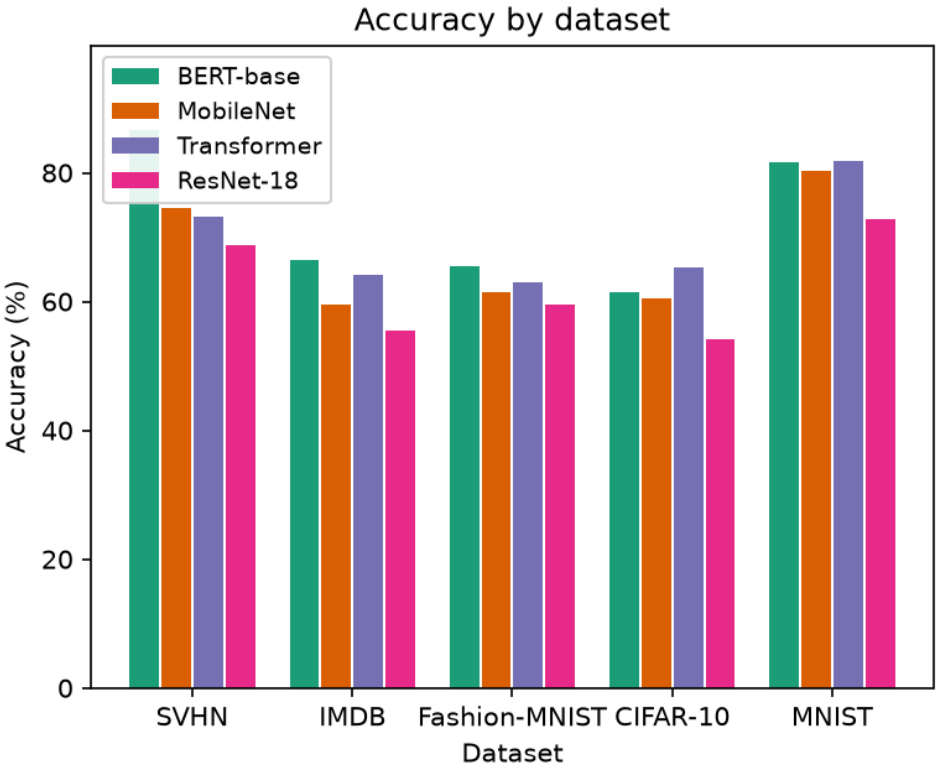


Figure 2: Accuracy by dataset for the models in this experiment.

Transformer outperforms BERT-base by 4 percentage points on CIFAR-10. BERT-base outperforms MobileNet by 21.3 percentage points on MNIST (Figure 2). MobileNet reaches over 75% accuracy on MNIST. As shown in Figure 2, BERT-base achieves 86.7% accuracy on SVHN.

3 Discussion

Random Forest reaches 89.4% accuracy on IMDB. We note that the training was stable for all models.